

No. of Functional Features Included in the Solution

Date	10 July 2026
Team ID	XXXXXX
Project Name	Raising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Secure Login	Authorized access control for the coordinator dashboard panel.	Admin Auth	Done	
2	API Ingestion	Fetches live weather, rainfall, and cloud metrics from external APIs.	Integration	Done	
3	Data Cleanse	Preprocesses raw weather fields to align with ML input layers.	Logic / BE	Done	
4	ML Inference	Runs cleaned metrics through .pkl weights to get threat levels.	ML Engine	Done	
5	Visual Alerts	Displays color-coded risk alerts (Low/Med/High) on the UI screen.	Frontend UI	Done	
6	Auto Logging	Automatically saves all prediction history runs into the dataset. Data	Data Store	Done	
7	History View	Allows users to query and review archived prediction audit records.	Backend/DB	Done	
8					

Feature Summary:

Metric	Count / Value
Total Features Planned	7
Total Features Implemented	7
Core / Must-Have Features	5

Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Interactive Flask Control Panel, Color-Coded Risk Alert Widgets
2	Backend / Logic	2	ML Inference Execution Engine, Weather Feature Engineering Module
3	Database / Storage	1	Automated Prediction Log Dataset System
4	API / Integration	1	Automated OpenWeatherMap Data Fetcher Client
5	Security / Authentication	1	Coordinator Secure Login Session Access Manager